Unit2 How markets work

2.1 demand

- definition: the quantity that consumers are willing and able to buy at a given price in a given period of time
- factor which determine the demand for good or service
  - price of good or service
    - price increases → lower quantity demanded
    - price increases → consumers cannot afford to buy as much → buy alternative products(relatively cheaper)
  - conditions of demand
    - income
      - definition: a flow of money, received by an economic agent over a period of time.
      - normal good: a product where demand increases if income rises.
      - inferior good: a product where demand decreases if income rises. (not always low quality → 有可能是cheaper alternatives → more likely to be bought by people with low incomes)
    - wealth
      - definition: a stock of assets owned by an economic agent at a point in time.
      - ↑ wealth → ↑ demand
    - the price of related goods and services
      - substitute goods or goods in competitive demand: alternative product- if the price of one good increases, so will the demand for another. They are in competitive demand.
      - complementary goods or goods in joint demand: products that are bought or used together- if the demand for one good increases, so will the demand for the other. They are in joint demand.
    - individual preference —— influenced by...
      - Advertising
        - persuade them to buy some products instead of others
        - successful advertising → ↑ demand
      - social and emotional factors
    - population —— more people live in a country → ↑ demand
- Movements along a demand curve and shifts in the demand curve
  - movement along the curve —— price of a product changes
  - shift of curve —— change of other conditions
    - to right → ↑ demand
    - to left → ↓ demand

ceteris paribus: nothing else change

2.2 elasticity of demand

- elasticity: definition: how much one variable changes in response to a change in other
- Price elasticity of demand(PED)
  - definition: the responsiveness of quantity demanded to a change in price
  - Elastic and inelastic demand
    - price and quantity demanded change in opposite directions → PED is negative
    - elastic: percentage change in price ＜ percentage change in quantity demanded
    - inelastic: percentage change in price ＞percentage change in quantity demanded
    - three extreme categories in PED
      - perfectly elastic
        -
        - any demand at a particular price but none at a higher price, PED is infinity
      - perfectly inelastic
        -
        - demand stays the same at all price, PED is 0
      - unit elasticity
        -
        - percentage change in demand ＝ percentage change in price, PED is -1
- Relationships between price elasticity of demand and firms' total revenue
  - demand is elastic, changes in price and total revenue move in opposite directions
  - demand is inelastic, changes in price and total revenue move in the same direction
  - demand is perfectly elastic, there is no demand at a higher price. price rises → total revenue falls to zero.
  - If demand is perfectly inelastic, demand stays the same at all prices → the percentage change in total revenue will be the same as the percentage change in price.
  - demand is of unit elasticity → for any change in price, total revenue will stay the same

total revenue: income from sales, calculated as price × quantity sold.

- Factors that influence PED
  - substitutes —— availability of substitutes ↑ → there are similar products that could be bought → more elastic
  - width of market —— wider market → more elastic
  - time —— longer time period → more elastic
  - percentage of income spent on the product —— large percentage of income is spent → more elastic(eg. car)
  - Necessities and luxuries?
    - usually necessities → inelastic
    - usually luxuries → elastic
- income elasticity of demand (YED)
  - $\text{YED} = \frac{\%\ \Delta \text{ in QD}}{\%\ \Delta \text{ in Y}}$ definition: the responsiveness of quantity demanded to a change in income
  - normal goods
    - YED is positive
    - income ↑ → likely that a higher percentage of income will be spent on holidays and lower percentage on food → luxury goods will usually have a higher values of YED than normal goods
  - inferior goods —— YED is negative
- Cross elasticity of demand(XED or CED)
  - $\text{XED} = \frac{\%\ \Delta \text{ in QD of A}}{\%\ \Delta \text{ in P of B}}$ definition: the responsiveness of quantity demanded of one good to a change in the price of another
  - subsititute goods
    - price of good B ↑ → demand for alternative A ↑ → XED is positive
    - ↑ XED → good substitutes → change in price of one good will have a large impact on the demand for the other good
  - complementary goods —— price of good B ↑ → demand of complementary good A ↓ → XED is negative
  - no relationship —— XED is 0

2.3 supply

- definition: the quantity that firms are able and willing to sell ar a given price in a given period of time
- Factors which determine the supply of a good or service
  - price of good or service —— ↑ price →firms are willing to offer more for sale → profit ↑ → ↑ quantity supplied
  - conditions of supply
    - costs of production —— costs ↑ → supply ↓
    - productivity
      - definition: how much a factor of production produces in a given time
      - labour productivity: how much a worker produces in a given time
      - ↑ productivity → cost per unit ↓ → ↑ supply
    - technical progress —— technical progress ↑ → supply ↑
    - number of firms in the market —— number of firms ↑ → supply ↑
    - weather and other events —— the output of agricultural goods can be affected by... and wars or strikes can also...
    - indirect taxes and subsidies
      - An indirect tax is a tax on spending. add costs → reduce supply
      - A subsidy is a payment to producers to reduce costs → increase supply

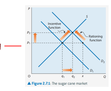
profit: total revenue minus total cost

- Movements along a supply curve and shifts in the supply curve —— 和demand curve一样

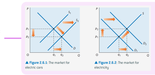
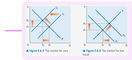
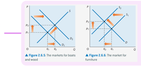
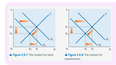
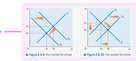
2.4 elasticity of supply

- definition: the responsiveness of quantiity suppplied to a change in price
- elastic and inelastic supply
- Perfectly elastic, perfectly inelastic and unit elasticity
  -
    - perfectly elastic
  -
    - perfectly inelastic
      - PES is 0
      - If a sports venue or theatre has a fixed number of seats, its supply will be perfectly inelastic.
  -
    - unit elasticity —— percentage change in supply＝ percentage change in price
- Factors that influence PES
  - availability of stock
  - spare capacity
  - impact on costs
  - nature of product
  - time
    - time ↑ →more elastic
    - Short run: the time period when at least one factor of production is fixed in suppply.
    - Long run: the time period when all factors of production are variable.
  - state of the economy

2.5 determination of market prices

- Factor markets and goods markets
  - Factor market: the market for buying and selling resources/ factors of production.
  - Goods/product market: the market for buying and selling goods and services.
- The rationing, incentive and signalling functions
  - Rationing function: the allocation of scarce resources and finished goods to those able and willing to pay
  - Signalling function: the idea that changes in price provide information to buyers and sellers in a market.
  - Incentive function: the idea that changes in price make producers or sellers of factors of production more or less likely to sell their goods or services.
- How the price mechanism is the way in which the basic economic problem is resolved in a market economy
  - 
    - Figure 2.7.1 The sugar cane market
  - 介绍一下这个图表，然后说一下price的三个functions是如何应用的，再说一下对别的市场的影响（比如factor market）
- Advantages and disadvantages of the price mechanism
  - Disadvantages of price mechanism
    - overproduced to meet people's preference→ harmful to the environment, underproduced or not produced
    - some people cannot afford even basic necessities
    - large firms may raise the price to earn more profit
    - resources may be wasted on advertising
  - advantages of price mechanism
    - choices of buyers and sellers of products are coordinated
    - match people's preference→ consumer sovereignty
    - freedom of choice
    - Competition will help to keep prices low→ resources will be used efficiently

2.6 interrelationship between markets

- Joint demand
  - price of electric cars ↓ →demand ↑ of electricity
  - 
    - Figure 2.6.1 The market for electric cars
    - Figure 2.6.2 The market for electricity
- Competitive demand
  - price of cars ↑ → demand for bus travel ↑
  - 
    - Figure 2.6.3 The market for cars
    - Figure 2.6.4 The market for bus travel
- Composite demand: a good which has more than one use
  - price of wood ↑ → cost of production of furniture ↑ →supply of furniture ↓
  - 
    - Figure 2.6.5 The markets for chairs and wood
    - Figure 2.6.6 The market for furniture
- Derived demand: where the demand for a factor of production results from the demand for the product it makes.
  - 
    - Figure 2.6.7 The market for chair
    - Figure 2.6.8 The market for chairs(factor)
- Joint supply: when the output of one good results in the output of another good.
  - wheat🌾extension in supply→ supply of straw
  - 
    - Figure 2.6.9 The market for wheat
    - Figure 2.6.10 The market for straw

2.7 How markets and prices allocate resources

- How the interaction of demand and supply determines equilibrium prices in a market economy
  - free market → price of a product is determined by the market forces of demand and supply
    - Market forces: demand and supply, which determine the price and quantity sold of products in a market economy.
  - At equilibrium, there is no tendency to change, unless there is a shift in either demand or supply.
    - Equilibrium price: the price where demand and supply for a product are equal.
    - Equilibrium quantity: the quantity where demand and supply for a product are equal.
- Difference between equilibrium and disequilibrium
  -
    - Figure 2.7.1 Equilibrium and disequilibrium
  - Disequilibrium: where there are forces at work that will lead to a change in a variable.
- Why excess demand and excess supply lead to changes in price
  - Shortage/excess demand: where demand is greater than supply at a given price.
    - at p1, the quantity demanded of pencils at y is ＞the quantity supplied of x → price will tend to rise towards equilibrium → this causes a contraction of demand and an extension of supply of pencils(because buyers will be less willing and able to buy pencils but suppliers will want to sell more, attracted by higher profits)
  - Surplus/excess supply: where supply is greater than demand at a given price.
    - at p2, the quantity supplied of pencils at w is higher than the quantity demanded of v→ price falls towards equilibrium→ this causes an extension of demand and a contraction of supply of pencils→ buyers are able and willing to buy more but there is less incentive for suppliers to sell as many
- Use of demand and supply diagrams to analyse causes of changes in market prices
  - An increase in demand
    -
      - Figure 2.7.2 An increase in demand
  - A decrease in demand
    -
      - Figure 2.7.3 A decrease in demand
  - An increase in supply
    -
      - Figure 2.7.4 An increase in supply
  - A decrease in supply
    -
      - Figure 2.7.5 A decrease in supply
- Significance of different price elasticities of demand and supply
  - If demand or supply is elastic, there will be a greater percentage change in quantity sold than price.
  - If demand or supply is inelastic, there will be a greater percentage change in price than quantity sold.
- Causes of fluctuations in commodity prices, including speculation
  - Commodities: primary products – raw materials, agricultural or mining products.
    -
      - Figure 2.8.1 Price change when demand and supply are inelastic
  - features of commodity
    - only can be produced under certain conditions (certain place or certain weather)
    - take a long time to produce
    - some have no substitutes
  - the quantity sold decreases a little bit, the price increases a lot
  - **Speculation:** the buying or selling of an asset to make a profit, by predicting that the price will rise or fall in the future.
  - **Asset:** something of value that is owned.
  - **Asset bubble:** when the price of a commodity or other asset increases quickly over a short period of time, usually due to speculation.

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